Labetalol

Mechanism of action

- Labetalol is a beta-blocker and an alpha-blocker. It antagonises (blocks) beta-1 receptors in the heart, causing a decrease in heart rate, cardiac output and blood pressure and it antagonises (blocks) alpha-1 receptors peripherally, causing vasodilation and a decrease in blood pressure.

Authorised practice levels

- ICPs and CCPs.
- Paramedics may only administer labetalol after seeking clinical advice.

Indications

- ✓ Control of hypertension prior to fibrinolytic therapy for STEMI.
- ✓ Control of hypertension during inter-hospital transfer for STEMI.
- ✓ Control of hypertension during inter-hospital transfer for stroke clot retrieval.
- ✓ Control of hypertension associated with autonomic dysreflexia.
- ✓ Control of hypertension associated with eclampsia.
- ✓ Control of hypertension during inter-hospital transfer for acute aortic dissection.

Contraindications

- ✗ Known severe allergy.
- ✗ Bradycardia. Labetalol will further reduce the heart rate.
- ✗ Hypotension. Labetalol will further reduce the blood pressure.

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Cautions

- 1st degree heart block. Labetalol may cause bradycardia.
- Known sick sinus syndrome without an internal pacemaker in place. Labetalol may cause bradycardia.
- Previous 2nd or 3 rd degree heart block without an internal pacemaker in place. Labetalol may cause worsening of heart block and bradycardia.
- Asthma or COPD. Labetalol may cause bronchospasm and should usually be withheld if the patient regularly takes bronchodilators.
- Heart failure. Labetalol will reduce cardiac output and may make heart failure worse.

Use in pregnancy or when breastfeeding

- Safe and should be administered when indicated.

Dosage

- Administer a 20 mg bolus IV every five minutes, as required, while setting up an infusion.
- Commence an IV infusion at 1 mg/kg/hour, up to a maximum of 150 mg/hour.
- Adjust the rate as required by 20 mg/hour every 10 minutes, titrating to effect.
- Administer additional boluses of 20 mg IV every five minutes as required.

The table below provides an overview of the blood pressure threshold for commencing treatment and the BP maintenance goal.

Indication	BP to commence treatment	BP maintenance goal
Prior to fibrinolytic therapy for STEMI	SBP > 180 mmHg, or DBP > 110 mmHg.	SBP 140-160 mmHg, and DBP < 110 mmHg.
IHT for STEMI	SBP > 180 mmHg, or DBP > 110 mmHg.	SBP 140-160 mmHg, and DBP < 110 mmHg.

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IHT for SCR	SBP > 180 mmHg, or DBP > 110 mmHg.	SBP 140-160 mmHg, and DBP < 110 mmHg.
Autonomic dysreflexia	SBP > 160 mmHg, or SBP > 40 mmHg above the patient's normal.	SBP 140-160 mmHg, and DBP < 110 mmHg.
Eclampsia	SBP > 160 mmHg, or DBP > 110 mmHg.	SBP 140-160 mmHg, and DBP < 110 mmHg.
IHT for acute aortic dissection	SBP > 120 mmHg.	SBP 100-120 mmHg.

Administration

- Draw up 100 mg undiluted and administer 20 mg as a bolus, before commencing an infusion via an infusion device.

Common adverse effects

- Hypotension.
- Bradycardia.
- Bronchospasm.

Usual onset of effect

- 2-3 minutes.

Usual duration of effect

- 1-2 hours.

Usual preparation

- Ampoule containing 100 mg in 20 ml.

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Pharmacokinetics

- Labetalol is metabolised in the liver.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- The blood pressure effect will be potentiated by other medicines that lower blood pressure. For example, GTN, antihypertensive medicines and amiodarone.
- The heart rate effects will be potentiated by other medicines that lower heart rate. For example, amiodarone and centrally acting calcium channel blockers such as diltiazem.

More Information

Version history

16 June 2026 1.1.0.1
Update: Indications and dosage updated.

9 June 2025 1.0.5.3
Update: Indications and dosage updated.

11 September 2023 1.0.4
Update: Dosage and additional information updated.